

Hands-on Activity 6.1	
Searching Techniques	
Course Code: CPE010	Program: Computer Engineering
Course Title: Data Structures and Algorithms	Date Performed: Sep 16, 2025
Section: CPE21S4	Date Submitted: Sep 16, 2025
Name(s): Punay, Heidee S.	Instructor: Engr. Jimlord Quejado
6. Output	

CODE

```
#include <iostream>
#include <cstdlib> // for generating random integers
#include <ctime>   // for time function

const int max_size = 50;

int main() {
    // Generate random values
    int dataset[max_size];

    srand(static_cast<unsigned int>(time(0)));

    // Fill the dataset with random values
    for(int i = 0; i < max_size; i++){
        dataset[i] = rand();
    }

    // Show the dataset's content
    for(int i = 0; i < max_size; i++){
        std::cout << dataset[i] << " ";
    }

    std::cout << std::endl;
    return 0;
}
```

OUTPUT	<pre> 2015042403 446822212 839925004 185521087 1874347174 1294407856 806933748 553370085 694746589 110584764 499476450 629238265 376772429 1303628224 461246792 2030382393 342113733 834280262 2124497543 1989469637 960139346 1912465853 1907496450 129348378 799586922 373649500 1865886657 1606046441 719143707 635471829 1719801830 586702462 1082294042 412243186 772223549 809157568 1706651042 1579157297 1362527654 253913983 1689742061 1862004104 883152249 2066514490 1018148680 1344399041 1949413236 1360262413 31195655 1926427131 === Code Execution Successful === </pre> 2nd Attempt: <pre> 817574669 1332811268 1761893429 1032871025 1389243479 1102481211 69738849 294585370 1882335438 1504663978 916886119 867691101 811201858 698582612 393840951 1717093962 417810461 233625764 1149153200 2113269272 1848967020 355670714 814375435 515381853 871857378 29828253 1177297743 1184528783 483199429 914220769 1665462383 1300774098 99548389 1279872164 186161475 1488791869 234869727 255900324 1783377239 2117205165 1760564302 552779710 837412618 424282512 1251362323 1231253570 2141376474 1669172784 1464879334 1143046026 === Code Execution Successful === </pre>
OBSERVATION	Based on the output, the code generates random values where the srand, it is used to seed the random number that everytime the program runs it shows a different sequence of random numbers.

7. Supplementary Activity

Problem 1:

```

#include <iostream>

int main() {
    int arr[] = {15, 18, 2, 19, 18,};
    int key = 18;
    int comparisons = 0;

    for (int i = 0; i < 5; ++i) {
        ++comparisons;
        if (arr[i] == key) {
            break;
        }
    }

    std::cout << "The element is " << key << "key " << comparisons << std::endl;
    return 0;
}

```

```
}
```

```
[Running] cd "d:\Heidee\DSA\" &&  
Comparisons is 18 The key is 2
```

The first thing that is done in the code is to declare and initialize the array that has 5 elements, where the elements are to be searched. The element or the number 18 is the number that should be found in an array. And was also asked to check the number of comparison.

8. Conclusion

To conclude, we were learning the searching techniques, which include binary search and linear search. In this activity, we were able to understand at least a bit of how searching techniques work in both the implementation of an array and a linked list.

9. Assessment Rubric